

Am. 3.

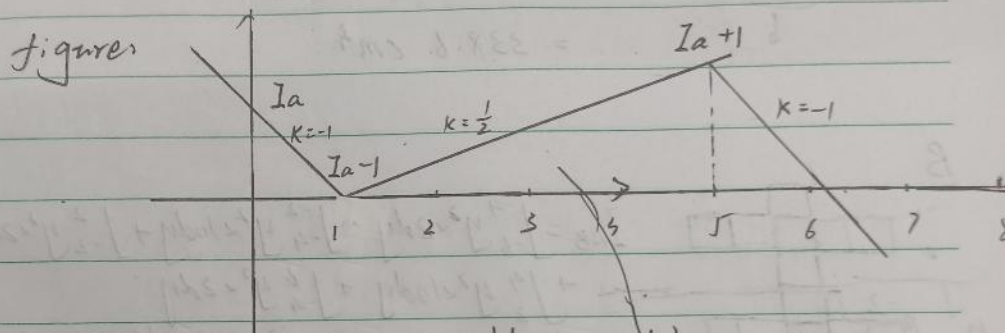
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Weather:

$$1. \begin{cases} \frac{dv}{dt} = f(v) - w + I_a \\ \frac{dw}{dt} = bv - w \end{cases} \Rightarrow \begin{cases} 0 = f(v) - w + I_a \\ 0 = bv - w \end{cases}$$

$$\text{nullclines: } \begin{cases} w = f(v) + I_a \\ w = bv \end{cases} \quad \text{and } f(v) = \begin{cases} -v, & v \leq 1 \\ \frac{1}{2}v - \frac{3}{2}, & 1 \leq v \leq 5 \\ 6 - v, & v \geq 5 \end{cases}$$

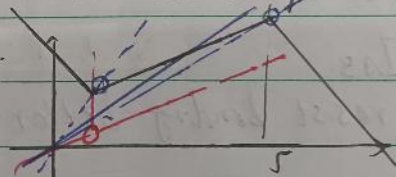


periodic: only unstable equilibrium point.

so $k > 0$. point at $k = \frac{1}{2}$.

no stable points.

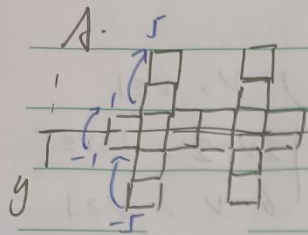
ideal situ.



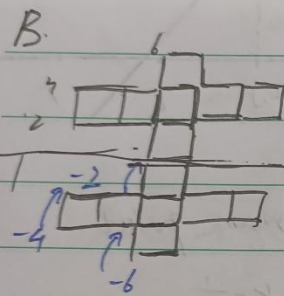
$$\begin{cases} I_a - 1 > \frac{1}{2} \\ \frac{I_a - 1}{1} > b > \frac{I_a + 1}{5} \end{cases}$$

$$\text{so } \begin{cases} I_a > \frac{3}{2} \\ I_a - 1 > b > \frac{I_a + 1}{5} \quad (b_{\min} = \frac{1}{2}) \end{cases}$$

2. Ans.



$$\begin{aligned}
 I_{AA} &= \int_{-6}^6 y^2 \cdot 4 \, dy + \int_{-12}^{-6} y^2 \cdot 12 \, dy + \int_6^{12} y^2 \cdot 4 \, dy \\
 &= \frac{4y^3}{3} \Big|_{-6}^6 + 4y^3 \Big|_{-12}^{-6} + \frac{4y^3}{3} \Big|_6^{12} \\
 &= 165.3 + 8 + 165.3 \\
 &= 338.6 \text{ cm}^4.
 \end{aligned}$$



$$\begin{aligned}
 I_{BB} &= \int_{-6}^6 y^2 \times 2 \, dy + \int_{-4}^{-2} y^2 \times 10 \, dy + \int_{-2}^2 y^2 \times 2 \, dy \\
 &\quad + \int_2^4 y^2 \times 10 \, dy + \int_4^6 y^2 \times 2 \, dy \\
 &= \frac{2y^3}{3} \Big|_{-6}^6 + \frac{10y^3}{3} \Big|_{-4}^{-2} + \frac{2y^3}{3} \Big|_{-2}^2 + \frac{10y^3}{3} \Big|_2^4 + \frac{2y^3}{3} \Big|_4^6 \\
 &= 101.3 + 186.7 + 10.7 + 186.7 + 101.3 \\
 &= 586.7 \text{ cm}^4.
 \end{aligned}$$

$$I_{AA} < I_{BB}$$

B will resist bending better.